

Silicon to graphene

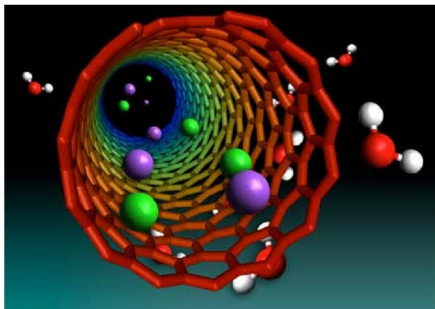
Silicon is the master base material of present-day electronics involving the transport of electrons and holes across the junction. However, there are growing complications related with the transport of these current carriers along with the increasing cost involved in miniaturisation of silicon-based electronics.

Efforts are under way to look for other base materials which are cheap but offer advantages similar to silicon. Nano-electronics is another area of research. It makes use of the wave nature of particles to generate electronic waves due to electron diffraction instead of electron diffusion. So the particles need not move and the energy is transferred in the form of waves.

Carbon nanosheets are considered the future of smaller, faster and cheaper electronics. High on the list is graphene. Ultra-strength, transparency (because of its thinness) and blisteringly fast electron conductivity make this nanomaterial promising for flexible displays and high-speed electronics. The theoretical mobility of electrons in pure graphene is 200 times that of silicon.

P.A.M. Dirac has established and highlighted the particle's wave nature at quantum level, and Dirac's equation explains the unique phenomena concerning the motion of an electron at nano level. It was found that an electron which seems to be moving slowly must actually have a very high-frequency oscillatory motion of small amplitude superposed on the regular motion. As a result of this oscillatory motion, the velocity of the electron at any time equals the velocity of light.

Carbon being available in plenty in its different forms like diamond and graphite has been under investigation as a base material for electronics for a long time. Diamond is very costly and hard but graphite—the material that gives pencils their marking ability—is cheap and flexible and could be the basis for a new class of nanometre-scale carbon-based electronic devices. These



Carbon nanotubes

devices will make use of the attractive properties of carbon nanotubes which can be produced using established microelectronics manufacturing techniques.

Recently, researchers have developed the foundation for circuitry and devices based on graphite. Graphene (electronic name given to graphite) is a condensed matter realisation of the Dirac electron relativistic wave equation at quantum level in two dimensions. Many of the unusual properties of relativistic fermions in quantum electrodynamics reappear in graphene.

The discovery of graphene, followed by observation of an anomalous quantum electronic transport phenomena in it, has triggered intense focus of research in this one-atom thin, two-dimensional system. The unique electronic band structure of graphene lattice provides a linear dispersion relation where the Fermi velocity replaces the role of the speed of light in usual Dirac's equation. Because of its high electronic mobility, structural flexibility and capability of being tuned from p-type to n-type doping by the application of a gate voltage, graphene is considered a potential breakthrough in carbon-based nano-electronics.

Using thin layers of graphite, researchers at the Georgia Institute of Technology in the United States, in collaboration with the Centre National de la Recherche Scientifique in France, have produced proof-of-principle transistors, loop devices and circuitry. Ultimately, the researchers hope to use graphene layers less than ten atoms thick as the basis for revolutionary electronic systems that manipulate electrons as waves rather than particles—much like photonic systems control light waves.

Graphene electronics

From graphene-based nano-electronics, scientists expect to make electronic devices of a kind that don't really have an analogue as in silicon-based electronics. The ultimate goal is integrated electronic structures that work on diffraction of electrons rather than diffusion of electrons. This will allow production of very small devices with very high efficiencies and low power consumption.

The graphene circuitry demonstrates high electron mobility—up to 25,000 cm² per volt-second—showing that electrons move with little scattering. Researchers have shown electronic coherence at near room temperature, and evidence of quantum interference effects. They expect to see ballistic